

Disproof of Conjecture 00000001021: The Minimum Order of a $(3, 6)$ -Regular Girth-12 Tanner Graph Is Not 132

earthking11

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Abstract

The conjecture asserts that the minimum number of vertices of a $(3, 6)$ -regular Tanner graph of girth 12 is 132, and is described as “computationally verifiable”. We show this is false. A Tanner graph is bipartite with a degree-3 vertex class (variable nodes) and a degree-6 vertex class (check nodes). If the girth is at least 12, then two distinct paths of lengths $i, j \leq 5$ from a fixed variable node cannot end at the same vertex, since they would close a cycle of length $i + j \leq 10 < 12$; hence the radius-5 ball around any variable node is a tree. Its BFS layers are forced to be

$$1, 3, 15, 30, 150, 300,$$

whose sum is 499. Therefore every such graph has at least 499 vertices, and in particular none has 132. Imposing the balance relation $3V = 6C$ of a $(3, 6)$ -regular bipartite graph strengthens the bound to 999. Inspecting the whole $(d_v, d_c) \in [2, 8]^2$, even-girth-4–16 parameter family, no Moore-type bound equals 132; the girth-12 lower bound is either 499 (tree-ball) or 999 (balanced), both far above 132.

1 Definitions

Definition 1 (Tanner graph). A *Tanner graph* is a finite bipartite graph $G = (X \cup Y, E)$ in which every edge joins a *variable node* in X to a *check node* in Y . It is (d_v, d_c) -regular if every vertex of X has degree d_v and every vertex of Y has degree d_c . Thus a $(3, 6)$ -regular Tanner graph has variable nodes of degree 3 and check nodes of degree 6.

Definition 2 (Girth). The *girth* of G is the length of a shortest cycle in G . Since G is bipartite, all cycles have even length.

We write $V = |X|$, $C = |Y|$, and $N = V + C$ for the total number of vertices. Because G is $(3, 6)$ -regular, double counting the edges gives

$$3V = 6C, \quad \text{equivalently} \quad V = 2C, \quad N = 3C = \frac{3}{2}V. \quad (1)$$

2 The collision lemma

Fix a variable node $v \in X$ and let $\text{dist}(v, \cdot)$ be graph distance in G . Let $B(v, r) = \{w : \text{dist}(v, w) \leq r\}$.

Lemma 3 (No collisions up to radius 5). *Let G be a Tanner graph of girth $g \geq 12$ and let v be any vertex. Write $r = \lfloor (g-1)/2 \rfloor$. Then the ball $B(v, r)$ is a tree. In particular, for $g = 12$, distinct paths from v of lengths $i, j \leq 5$ have distinct endpoints.*

Proof. Suppose two distinct paths P_1, P_2 from v of lengths i and j share their terminal vertex w . Walking along P_1 from v to w and back along P_2 from w to v yields a closed walk of length $i + j$. If P_1, P_2 are distinct, this closed walk contains a cycle; its length is at most $i + j$ and, being a cycle in a bipartite graph, it is even and hence at least 2. More precisely, the symmetric difference of the two paths contains a cycle of length $\leq i + j$. If $i, j \leq 5$ then $i + j \leq 10 < 12 \leq g$, contradicting the definition of girth. Hence no two such paths collide.

It remains to see that the whole ball, not just its endpoints, is a tree.

1. *No edges inside a layer.* In a bipartite graph an edge joining two vertices at the same distance from v would create an odd cycle, impossible.
2. *No two layers meet.* Distance is single-valued: a vertex cannot have two different distances from v .
3. *No cross edges.* An edge between a vertex at depth d and a vertex at depth d' with $|d - d'| \neq 1$ would shortcut the BFS, forcing $\text{dist}(v, \cdot) \leq \min(d, d') + 1 < \max(d, d')$, contradicting single-valuedness of the other depth.

Combining these facts, the edges of $B(v, r)$ are exactly the tree edges of BFS, with no extra edge possible among depths $0, \dots, r$ when $r \leq g/2 - 1$. For $g \geq 12$ we have $\lfloor (g-1)/2 \rfloor \geq 5$, so the radius-5 ball is a tree. $\square$

Remark 4. The same argument shows that $B(v, 5)$ is a subgraph of G with 499 distinct vertices; it need not be an induced subgraph, but the collision argument above already rules out every extra edge that could occur inside the ball at these depths.

3 Forced BFS layers and the 499 bound

Let G be $(3, 6)$ -regular of girth $g \geq 12$ and let v be a variable node. By Lemma 3, the radius-5 ball around v is a tree, so the BFS layers are forced by the degrees, with one edge from each non-root vertex used by its parent.

depth d	vertex class at depth d	children per vertex	layer size	cumulative
0	variable (root, no parent)	3	1	1
1	check (6 neighbours, 1 parent)	5	3	4
2	variable (3 neighbours, 1 parent)	2	15	19
3	check	5	30	49
4	variable	2	150	199
5	check	—	300	499

The layer sizes are obtained by the recurrence

$$\begin{aligned} \ell_0 &= 1, & \ell_1 &= 3\ell_0 = 3, & \ell_2 &= 5\ell_1 = 15, \\ \ell_3 &= 2\ell_2 = 30, & \ell_4 &= 5\ell_3 = 150, & \ell_5 &= 2\ell_4 = 300. \end{aligned}$$

Theorem 5. *Every $(3, 6)$ -regular Tanner graph of girth 12 has at least 499 vertices:*

$$|B(v, 5)| = 1 + 3 + 15 + 30 + 150 + 300 = 499 \leq N.$$

Consequently the minimum number of vertices of such a graph is not 132, and indeed no such graph has 132 vertices.

Proof. Lemma 3 makes the six layers pairwise disjoint, and the layer sizes are forced as above. Their sum is 499, all inside G . Since $132 < 499$, the claimed value cannot occur. $\square$

Note that 499 is only a lower bound on the true minimum; the disproof needs nothing more.

4 The balanced variant: 999

The count above ignores the balance relation (1), which a Tanner graph must satisfy. Splitting the forced ball by vertex class: the variable nodes at even depth number $1 + 15 + 150 = 166$, and the check nodes at odd depth number $3 + 30 + 300 = 333$. Hence

$$V \geq 166, \quad C \geq 333.$$

By (1), $V = 2C$, so $C \geq 333$ already gives $V \geq 666$ and

$$N = V + C = 3C \geq 3 \cdot 333 = 999.$$

Equivalently, among the 333 forced check nodes each is adjacent only to variable nodes, and a $(3, 6)$ -regular graph needs twice as many variable as check nodes; the check-side count is the binding constraint.

Theorem 6. *Every $(3, 6)$ -regular Tanner graph of girth 12 has at least 999 vertices.*

Both bounds (499 from the tree-ball alone, 999 after balancing) are far from 132.

5 No convention in the parameter family yields 132

For general (d_v, d_c) -regular Tanner graphs of even girth $2r$, the same argument forces the BFS layers

$$\ell_0 = 1, \quad \ell_{i+1} = \ell_i b_i,$$

where the branching is

$$b_0 = d_v, \quad b_i = d_c - 1 \quad (i \text{ odd}), \quad b_i = d_v - 1 \quad (i > 0 \text{ even}).$$

The tree-ball lower bound is $\sum_{i=0}^r \ell_i$. Combining it with the balance relation $d_v V = d_c C$ gives a balanced lower bound on $N = V + C$, obtained by requiring $V \geq \sum_{i \text{ even}} \ell_i$ and $C = \frac{d_v}{d_c} V \geq \sum_{i \text{ odd}} \ell_i$.

Scanning all $2 \leq d_v, d_c \leq 8$ and all even girths $4, 6, \dots, 16$ (the verification is in `reproduce.py`) yields *no* parameter choice whose tree-ball or balanced bound equals 132.

For the specific family $(d_v, d_c) = (3, 6)$ the rigorous balanced bounds are:

girth	tree-ball lower bound	balanced lower bound
4	4	9
6	19	24
8	49	99
10	199	249
12	499	999
14	1999	2499
16	4999	9999

The value 132 matches none of these: the rigorous bounds are 24 for girth 6, 99 for girth 8, 249 for girth 10, and 999 for girth 12. Even the raw tree-ball bound for girth 12 is 499. There is therefore no reading of “girth 12” under which 132 is the minimum: 132 is simply too small, and the claimed “minimal graph counting” is incorrect.

6 Conclusion

Conjecture 00000001021 is false. The radius-5 ball around a degree-3 variable node in a $(3, 6)$ -regular girth-12 Tanner graph is a tree with $1 + 3 + 15 + 30 + 150 + 300 = 499$ vertices, so every such graph has at least 499 vertices; the balanced bound is 999. No (d_v, d_c) -regular girth-12 graph, and no member of the scanned parameter family, can have only 132 vertices.